

Expansion Planning of Active Distribution Networks With Multiple Distributed Energy Resources and EV Sharing System

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Abstract—The ever-increasing energy demand and high penetration rate of distributed renewable generation brings new challenges to the planning of power distribution networks. This paper proposes an expansion planning model for distribution networks by considering multiple types of energy resources in distribution side, including shared electric vehicle (SEV) charging stations, solar-based distributed generation sources, and battery energy storage systems. The operational characteristics of SEV are considered and modeled. The proposed planning model aims to minimize the weighted sum of network investment cost, energy losses, and queue waiting time of SEVs. A stochastic scenario generation method is introduced to address the stochastic feature of SEVs' driving behaviors. Numerical studies are tested on the systems with 54-node distribution network and 25-node traffic network.

Index Terms—Active distribution network, shared electric vehicles, distributed energy resources, traffic network.

NOMENCLATURE

Sets

Ω^D	Set of nodes in distribution system;
T	Set of time intervals;
H	Total load level number;
$\Omega_{old}^S, \Omega_{new}^S$	Set of substations with reinforcement and construction;
$\Omega_{old}^L, \Omega_{new}^L$	Set of feeders with reinforcement and construction;
Ω^C, Ω^{BE}	Set of candidates for Shared Electric Vehicle Charging Stations (SEVCSs) and Battery Energy Storage Systems (BESSs);

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Ω^{SDG}

Set of candidates of solar based distributed generation (SDG).

Parameters

ω	Power consumption of SEV battery per 100 km;
η	Charging efficiency of SEV;
β	Penetration rate of SEV;
S_{soc0}, S_{soc1}	Initial and end state of charge of SEV;
$P_{k,t}^{CH}$	The k th SEVCS charging demand in time t ;
$P_{i,t}^{SL}$	Demand of slow charging at node i in time t ;
$P_{w,t}^{SDG}$	Solar power output of w th SDG in time t ;
$P_{i,t}^D, Q_{i,t}^D$	The demand for active and reactive power at node i in time t ;
l_{ij}, z_k	Length of feeder ij and size of k th SEVCS;
$E_w^{SDG}, E_{m,t}^{BE}$	Energy capacity of w th SDG and m th BESS;
$f_s^{\max}$	The upper limit of power flow;
$z_s^{\max}, z_s^{\min}$	Upper and lower limits for SEVCSs size;
$I_{\max}, I_{\min}$	Current upper and lower limits of feeder ij ;
$E_{\max}^{BE}, E_{\min}^{BE}$	Upper and lower limits for BESSs size;
$P_{bat,\max}$	BESS rated power;
$P_{ch,m,t}, P_{dis,m,t}$	Charging and discharging power of m th BESS in time t ;
η_{ch}, η_{dis}	The efficiency of BESS charging/discharging;
$P_{rated}^{SDG}, E_{\max}^{SDG}$	The rated power and upper limit of SDG capacity;
m^D, m^S	The nodes and substations number in distribution system;
$S_{i,old}^S, S_{i,new}^S$	Apparent power capacity of i th substation;
S_{ij}^L	Apparent power capacity for feeder ij ;
g_{ij}, b_{ij}	Conductance and susceptance of feeder ij ;
G_{ij}, B_{ij}	Real and imaginary parts of node admittance matrix;
$F_{SL}(c), F_{CH}(c)$	Slow and fast charging speed when current battery level is c ;
$U_{i,\min}^{\min}, U_{i,\max}^{\max}$	Voltage limits of node i ;
$W_{ij,old}^L, W_{ij,new}^L$	Maintenance cost of feeder ij and i th substation;
$W_{i,old}^S, W_{i,new}^S$	Maintenance cost of i th substation;
W_k^C, W_m^{BE}	Maintenance cost of k th SEVC, m th BESS
$W_{w,t}^{SDG}$	and w th SDG;
$c_{ij,old}^L, c_{ij,new}^L$	The reinforcement and construction capital cost of feeder ij ;

$c_{i,old}^S, c_{i,new}^S$	The reinforcement and construction capital cost of i th substation;
$c_k^{I,CH}, c_m^{I,BE}, c_w^{I,SDG}$	Capital cost of k th SEVC, m th BESS and w th SDG;
$c_k^{CH}, c_m^{BE}, c_w^{SDG}$	Unit fixed cost proportional to size of k th SEVC, m th BESS and w th SDG;
b, c^E	The investment rate and energy cost;
δ	Battery degradation coefficient of BESS;
$\alpha_{old}^L, \alpha_{new}^L$	Capital recovery factor for feeder reinforcement and construction;
$\alpha_{old}^S, \alpha_{new}^S$	Capital recovery factor for substation reinforcement and construction;
$\alpha^c, \alpha^{BE}, \alpha^{SDG}$	Capital recovery factors for SEVC, BESS, and SDG;
$f_q^{rs}(t)$	Size of path q flow which connecting rs in time t .

I. INTRODUCTION

SINCE twenty-first century, power delivery side has been experiencing a profound transition from “passive” to “active” operation pattern. In 2009, the concept of “Active Distribution Network (ADN)” was introduced by the International Council on Large Electric Systems (CIGRE). Generally, an ADN refers to a distribution system that integrates a combination of Distributed Energy Resources (DERs) including Electric Vehicles (EVs), Battery Energy Storage Systems (BESSs), Renewable Energy Sources (RESs), and so on. In this context, optimal planning of ADNs is recognized as a critical and fundamental task in modern grids’ operations.

In literature, a great variety of models has been proposed for ADN expansion planning with integration of RESs. Reference [1] proposes a multistage planning model for distribution network expansion, in which investments for both the distribution network and Distributed Generation (DG) are considered. Reference [2] proposes a two-step algorithm that incorporates renewable uncertainty and reliability into a multi-stage distribution network expansion model. A Mixed Integer Linear Programming (MILP) model is introduced in [3] to address the sizing and siting problem of wind-based DGs (WDGs) and solar-based DGs (SDGs) in a radial distribution system; the objective is to minimize the cost of system investment and operation. In addition to distributed RESs, BESSs are also commonly considered in ADNs’ planning and operation. In [4], optimal sizes of BESSs are determined in each bus, based on a cost-benefit analysis model that considers voltage regulation and peak load shaving. Reference [5] propose a BESS planning model in ADN by considering the BESS investment cost, system maintenance cost, and deferring facility investment.

Besides BESSs and DGs, some researches investigate ADN expansion planning with integration of EVs. Adoption of EV can reduce air pollutant emission and contribute to more pronounced use of renewable energy. Reference [6] proposes an integrated planning framework to effectively accommodate EVs in a distribution system. The method incorporates various investment options available for the utility, including distributed generation and network reinforcement.

A coordinated optimal planning model for EV Charging Stations (EVCSSs) and DGs is proposed in [7]; the model coordinates several planning objectives, including system investment cost, system reliability, power loss, and voltage stability.

The fact that multiple types of energy resources (i.e., EVCSSs, BESSs, and DRERs) are deeply coupled in ADNs implies that the sizing and sitting of different kinds of energy resources should be considered simultaneously in ADN’s expansion planning. Very limited research can be found to address such joint planning problem. Reference [8] proposes an optimization model for selecting the size and location of EVCSSs, BESSs, and DERs. However, the optimization model in [8] is based on a fixed distribution network structure and does not consider the future network expansion. Additionally, the stochastic nature of EV driving behaviours is also an important factor. A scenario-based stochastic program is presented in [2] to model the uncertainty of EV charging demands. In [9], uncertainties of EV charging loads are modelled for different types of charging stations. The Monte Carlo Sampling (MCS) method has been applied in [9] to construct the joint probability distribution scenarios for stochastic variables. The major challenge of the MCS-based approach is that to achieve a satisfactory probabilistic distribution approximation accuracy, a large number of samples needs to be generated, which could be highly computational expensive.

Furthermore, there is no research considering EV sharing systems in distribution network’s planning, which are increasingly prevalent in many countries. With car sharing services, travellers can access a fleet of spatially distributed vehicles at any time. As a result, significant cost savings can be achieved since users do not need to purchase and maintain personal vehicles. In this context, a reasonable sitting and sizing scheme for Shared EV Charging Stations (SEVCSs) is important when planning a distribution network.

In this study, a planning model for joint ADN is proposed by coordinating multiple types of energy resources. Our key contributions are summarized as follows:

(1) To the authors’ knowledge, this is the first research that considers EV sharing systems in the long-term joint ADN planning problem. The proposed ADN planning model simultaneously determines the size and location of heterogeneous energy resources in ADN (i.e., SEVCSs, BESSs, and SDGs), as well as the distribution network expansion scheme.

(2) This study proposes a scenario analysis method that fully considers the stochastic feature of SEV charging loads. The method is based on the Wasserstein metric and K-medoids approach; it not only improves the approximation precision of the scenario set, but also is highly computational efficient due to its parallel architecture.

This paper is organized as follows. In Section II, the stochastic modelling methodology for SEV load demand is presented. Section III presents the proposed joint AND planning model. Section IV presents the optimization approach and simulation results. Section V gives the conclusion and future work.

II. STOCHASTIC MODELLING OF SEV LOAD DEMAND

In this study, slow charging posts are assumed to be installed in each node of the distribution network as the primary charging facility for shared EVs. SEVCSs provide complementary support to fuel SEVs with fast charging; all SEVs are recharged once they arrive the SEVCS. The number of slow charging posts is proportional to the daily charging demand of each node that can be calculated following [10]. For fast charging behaviors in SEVCSs, the Wasserstein distance method, which has been proven to be useful and applied in different fields such as signal processing and machine-learning [11], is used to model the uncertainty of SEV's start charging time and charging duration. Afterwards, an improved K-medoids clustering scenario reduction method is applied to reduce the scenario number of the joint uncertainty states.

After above procedures, the daily SEV charging profile can be obtained. The start charging time of a SEV is the time when it returns to the charging station and connects to the charging port. The time of shared cars returning to charging stations at different time intervals is from the Car2Go data in Austin, Texas [12], which can be fit by using the sub-normal distribution shown in Eq. (1). This dataset contains one-month time stamp records of all SEVs in the service region of Car2Go company in Austin. The dataset includes following items: (a) when and where the user picks-up and returns their EVs, (b) the EV's battery level, and (c) the charging status of each car.

$$f_{ss}(t_{ss}) = \begin{cases} \frac{1}{\sigma_s \sqrt{2\pi}} \exp\left(\frac{-(t_{ss}-\mu_s)^2}{2\sigma_s^2}\right) & \mu_s - 12 < t_{ss} \leq 24 \\ \frac{1}{\sigma_s \sqrt{2\pi}} \exp\left(\frac{-(t_{ss}+24-\mu_s)^2}{2\sigma_s^2}\right) & 0 \leq t_{ss} \leq \mu_s - 12 \end{cases} \quad (1)$$

where t_{ss} denotes the SEV start charging time; μ_s and σ_s represent the expectation and standard deviation of t_{ss} , respectively. The SEV's start charging time can be any time in the day, starting from midnight ($t_{ss} = 0$). In order to find the most approximate quantiles for the start charging time's probability distribution function, the discretization process of Eq. (1) can be obtained according to the Wasserstein distance metric defined below.

Definition of Wasserstein distance metric: The distance between two known distributions p_1 and p_2 can be defined as [13]:

$$W_r(p_1, p_2; \pi) = \int \pi[p_1(x), p_2(x)]^r dx \quad (2)$$

where $\pi(p_1, p_2)$ represents the measurement of the distance; the order is denoted by r . Obtaining the most identical discretised probability distribution for the given continuous probability function is equivalent of minimizing the distance (W_r) between p_1 and p_2 . Supposing a continuous probability distribution $f(x)$ is estimated based on a discretised distribution with Q quantiles, the z_q ($q = 1, 2, \dots, Q$) indicates the optimised quantile can be obtained as follows:

$$\int_{-\infty}^{z_q} f(x)^{1/(1+r)} dx = \frac{2q-1}{2Q} \int_{-\infty}^{\infty} f(x)^{1/(1+r)} dx \quad (3)$$

The corresponding probability of each optimised quantile is calculated as:

$$\begin{cases} P^q = \int_{\frac{z_{q-1}+z_q}{2}}^{\frac{z_q+z_{q+1}}{2}} f(x) dx, & q = 1, \dots, Q \\ P^0 = \int_{z_0}^{\frac{z_1+z_2}{2}} f(x) dx \\ P^{Q+1} = \int_{\frac{z_{Q-1}+z_Q}{2}}^{z_{Q+1}} f(x) dx \end{cases} \quad (4)$$

By substituting the Eq. (1) into the left side of Eq. (3) and using the cumulative density function of normal distribution to simplify the integral, following equations can be derived:

a) When $0 \leq t_{ss} \leq \mu_s - 12$:

$$\int_0^{t_{ss}} f_{ss}(t)^{1/(1+r)} dt = \frac{1}{2} \sqrt{1+r} (\sqrt{2\pi}\sigma_s)^{r/(1+r)} \times \left[\operatorname{erf}\left(\frac{t_{ss} - (\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right) - \operatorname{erf}\left(\frac{-(\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right) \right] \quad (5)$$

b) When $\mu_s - 12 < t_{ss} \leq 24$:

$$\int_0^{t_{ss}} f_{ss}(t)^{1/(1+r)} dt = \frac{1}{2} \sqrt{1+r} (\sqrt{2\pi}\sigma_s)^{r/(1+r)} \times \left[2\operatorname{erf}\left(\frac{12}{\sqrt{2(1+r)}\sigma_s}\right) - \operatorname{erf}\left(\frac{-(\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right) + \operatorname{erf}\left(\frac{t_{ss} - \mu_s}{\sqrt{2(1+r)}\sigma_s}\right) \right] \quad (6)$$

The definition of the error function (erf) in Eqs. (5) and (6) is based on [14]. By substitute integrals in Eqs. (5) and (6) into Eq. (3), Eqs. (7) and (9) are obtained. Then by solving Eqs. (7) and (9), the optimised quantile z_q can be obtained from Eqs. (8) and (10). The probability of each optimised quantile P^q is calculated based on Eq. (4).

c) When $q < Q(\mu_s - 12)/24$:

$$\begin{aligned} & \operatorname{erf}\left(\frac{z_q - (\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right) - \operatorname{erf}\left(\frac{-(\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right) \\ &= \frac{2q-1}{Q} \operatorname{erf}\left(\frac{12}{\sqrt{2(1+r)}\sigma_s}\right) \\ z_q &= \sqrt{2(1+r)}\sigma_s \operatorname{erf}^{-1}\left[\frac{2q-1}{Q} \operatorname{erf}\left(\frac{12}{\sqrt{2(1+r)}\sigma_s}\right) + \operatorname{erf}\left(\frac{-(\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right)\right] + \mu_s - 24 \end{aligned} \quad (7)$$

d) When $q > Q(\mu_s - 12)/24$:

$$\begin{aligned} & \frac{2q-1-2Q}{Q} \operatorname{erf}\left(\frac{12}{\sqrt{2(1+r)}\sigma_s}\right) + \operatorname{erf}\left(\frac{-(\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right) \\ &= \operatorname{erf}\left(\frac{z_q - \mu_s}{\sqrt{2(1+r)}\sigma_s}\right) \\ z_q &= \sqrt{2(1+r)}\sigma_s \operatorname{erf}^{-1}\left[\frac{2q-1-2Q}{Q} \operatorname{erf}\left(\frac{12}{\sqrt{2(1+r)}\sigma_s}\right) + \operatorname{erf}\left(\frac{-(\mu_s - 24)}{\sqrt{2(1+r)}\sigma_s}\right)\right] + \mu_s \end{aligned} \quad (9)$$

For charging duration, we firstly analyse the SEV daily mileage. Based on origin-destination distances of the

Car2Go Trips data, the SEV mileage follows the lognormal distribution [12], [15]:

$$f_d(d) = \frac{1}{x\sigma_d\sqrt{2\pi}} \exp\left(-\frac{(\ln d - \mu_d)^2}{2\sigma_d^2}\right) \quad (11)$$

where d represents the distance traveled by SEV; μ_d and σ_d are expectation and standard deviation of SEV mileage, respectively. Eq. (12) is applied to calculate the charging duration:

$$t_{cs} = \frac{d \bullet \omega}{100P^{CH}\eta} \quad (12)$$

where t_{cs} is charging duration; P^{CH} is charging power of SEVCS. Unlike the start charging time model in Eq. (1), substituting the probability density function of charging duration into Eq. (3) gives no analytical solution to the discrete quantiles z_q . Hence, we numerically compute the discretion quantiles z_q , such that Eq. (3) holds. This problem can be transformed into following optimization problem:

$$z_q = \arg \min_{z_q^*} \int_{-\infty}^{z_q^*} f_{cs}(t_{cs})^{1/(1+r)} dt_{cs} - \int_{-\infty}^{\infty} f_{cs}(t_{cs})^{1/(1+r)} dt_{cs} \quad (13)$$

To solve model (13), a robust optimization routine is needed to minimize the term $\int_{-\infty}^{z_q^*} f_{cs}(t_{cs})^{1/(1+r)} dx - \int_{-\infty}^{\infty} f_{cs}(t_{cs})^{1/(1+r)} dx$, and a computationally efficient numerical integration routine is needed to approximate the two definite integrals. In this study, the Runge-Kutta algorithm [16] is used as the numerical integration routine; the Levenberg-Marquardt (LM) algorithm [17] is used as the optimization routine. LM outperforms gradient descent by implementing a damping term [18] and it converges much faster than gradient descent [19]. Discrete quantile z_q then can be computed numerically. Once all discrete quantiles of start charging time and charging duration are obtained, these states can be paired through the Cartesian product and the joint uncertainty states can be constructed for each hour. Denoting $\xi = 1$ when SEV is charged and $\xi = 0$ when it is fully charged or not charged; denoting the occurrence probability of each joint state is ρ_r . Then, the charging probability of each joint state satisfies following conditions:

$$P(\xi = 1) = 1 - \rho_r(t_{ss} > t, t_{ss} + t_{cs} \leq t + 24) - \rho_r(t_{ss} + t_{cs} \leq t) \quad (14)$$

$$P(\xi = 0) = \rho_r(t_{ss} > t, t_{ss} + t_{cs} \leq t + 24) + \rho_r(t_{ss} + t_{cs} \leq t) \quad (15)$$

To generate the scenario set of charging probability over 24 hours, multi-time period scenarios are required to be generated based on the state of charging probability of each hour. However, with increase of T , the number of generated scenarios will be increased exponentially, leading to dimensionality curse. To tackle this, an improved scenario clustering reduction method is used. Firstly, T is categorized into different sub-intervals, Then, the K-medoids clustering method [20] is used to reduce the each sub-interval scenarios into K scenarios, then all K scenarios of neighboring sub-intervals are fused by the

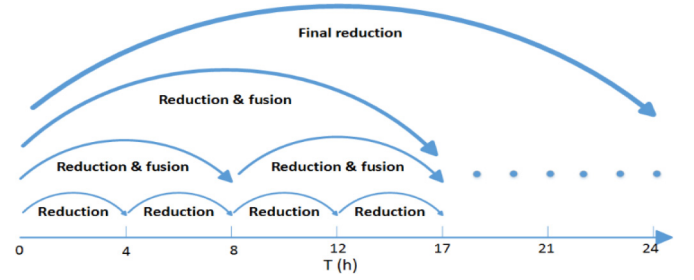


Fig. 1. Improved K-medoids Scenario Reduction approach.

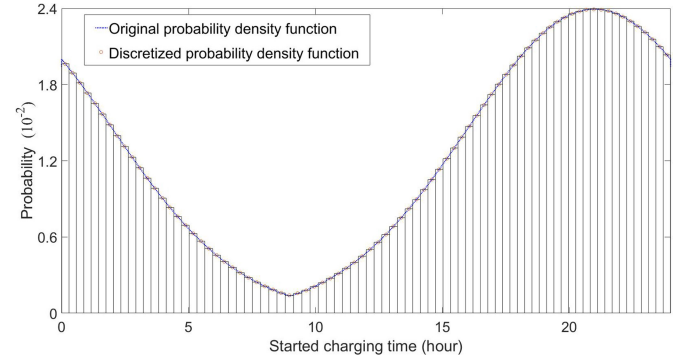


Fig. 2. Discretized probabilistic modeling of started charging time.

Cartesian product approach based on [21]. Finally, this reduction and fusion procedure repeats until the whole time horizon is covered, as illustrated in Fig. 1. Once the scenario set of SEV charging load of the whole time horizon is obtained, the SEV charging demand of the k th SEVCS in each time interval can be derived as:

$$P_{k,t}^{CH} = \frac{1}{n_{re}} z_k \sum_{n=1}^{n_{re}} \sum_{t=1}^{24} (\xi_t \cdot N_{EV} \cdot P_{fast}) p_n^{re} \quad (16)$$

where z_k represents the ratio of k th SEVCS charging demand and the total charging demand obtained following [10]. p_n^{re} and n_{re} are the probability of each reduced scenario and the number of reduced scenarios, respectively.

III. EXPANSION PLANNING MODEL OF JOINT ACTIVE DISTRIBUTION NETWORKS

The stochastic expansion planning model of joint ADN is introduced in this section, which aims to minimize the total annual investment cost, energy loss, and SEV waiting cost.

$$\min F = f_L + f_S + f_{CH} + f_{BE} + f_{SDG} + f_{loss} + f_w \quad (17)$$

$$f_L = \sum_{(ij) \in \Omega_{old}^L} x_{ij} (\alpha_{old}^L c_{ij,old}^L l_{ij} + W_{ij,old}^L) + \sum_{(ij) \in \Omega_{new}^L} x_{ij} (\alpha_{new}^L c_{ij,new}^L l_{ij} + W_{ij,new}^L) \quad (18)$$

$$f_S = \sum_{i \in \Omega_{old}^S} y_i (\alpha_{old}^S c_{i,old}^S + W_{i,old}^S) + \sum_{i \in \Omega_{new}^S} y_i (\alpha_{new}^S c_{i,new}^S + W_{i,new}^S) \quad (19)$$

TABLE I
VARIABLES OVERVIEW

Variable	Definition
x_{ij}, y_i	Binary decision variables of feeder ij and i th substation
u_k, e_m, g_w	Binary decision variables of SEVCS, BESS, SDG
$U_{i,t}, U_{j,t}$	Voltage magnitude of bus i and bus j in time t
I_{ij}	Current magnitude of feeder ij
$\theta_{ij,t}$	Phase angle deviation of branch ij in time t
$P_{ij,t}, Q_{ij,t}$	Active and reactive power of feeder ij in time t
$P_{i,n}^s, Q_{i,n}^s$	Active and reactive power provided by substation at node i in time t

$$f_{CH} = \sum_{k \in \Omega^C} u_k \left[\alpha^C \left(c_k^{I,CH} + c_k^{CH} z_k \right) + W_k^C \right] \quad (20)$$

$$f_{BE} = \sum_{m \in \Omega^{BE}} e_m \left[\alpha^{BE} \left(c_m^{I,BE} + c_m^{BE} E_m^{BE} \right) + W_m^{BE} \right] + \sum_{t \in T} \sum_{m \in \Omega^{BE}} P_{ch,m,t} |P_{dis,m,t}| \delta \quad (21)$$

$$f_{SDG} = \sum_{w \in \Omega^{WDG}} g_w \left[\alpha^{SDG} \left(c_w^{I,SDG} + c_w^{SDG} E_w^{SDG} \right) + W_w^{SDG} \right] \quad (22)$$

$$f_{loss} = 365 \cdot c^E \sum_{(ij) \in \Omega^L} \sum_{t \in T} \left[g_{ij} x_{ij} \left(U_{i,t}^2 + U_{j,t}^2 - 2U_{i,t}U_{j,t} \cos \theta_{ij,t} \right) \right] \quad (23)$$

$$f_w = 365 \sum_{r \in \Omega^T} \sum_{r \neq s \in \Omega^T} v \sum_{t=1}^T q_{rs} \Delta T \quad (24)$$

where

$$\alpha = \frac{i(1+b)^{n^{life}}}{(1+b)^{n^{life}} - 1} \quad (25)$$

$$q_{rs}(t) = q_{rs}(t-1) + \sum_{q \in Q_{rs}} \beta q_q^{rs}(t) - d_{rs}(c_{rs}, t) t \in [1, T] \quad (26)$$

$$d_{rs}(c, t) = \begin{cases} d_{rs}(c_{rs}, t) & \forall c \in [0, c_{rs}] \\ \min \left\{ \begin{aligned} &\Delta T \psi_{rs}, \\ &\left\{ I_r(c, t) + A_r(c, t) - \sum_{k \in \Omega^T} d_{rk}(c, t) \right\}, \\ &\left(\sum_{q \in Q_{rs}} \beta q_q^{rs}(t) + q_{rs}(t-1) \right) \end{aligned} \right\}, & \forall r \neq s \in \Omega^T, t \in [1, T], c \in [c_{rs}, C] \end{cases} \quad (27)$$

$$A_r(c, t) = \begin{cases} \sum_{s \neq r \in \Omega^T} d_{sr}(c + c_{rs}, t - t_{rs}), & c \in [0, C - c_{rs}] \\ 0, & c \in [C - c_{rs}, C] \end{cases} \quad (28)$$

$$I_r(c, t) = I_r(c - F_{SL}(c) - \mu_k F_{CH}(c), t-1) + A_r(c - F_{SL}(c) - \mu_k F_{CH}(c), t-1) - \sum_{r \neq s} d_{rs}(c - F_{SL}(c) - \mu_k F_{CH}(c), t-1) \quad (29)$$

The objective function Eq. (17) includes: (1) the total annual cost of feeders (f_L), substations (f_S), SEVCs (f_{CH}), BESSs (f_{BE}), and SDGs (f_{SDG}); (2) energy losses cost (f_{loss}); and (3) waiting cost of SEV (f_w). All variables in the proposed model are shown in table I. Eqs. (18) and (19) calculate

the reinforcing/constructing cost of feeders and substations. Eq. (20) calculates the SEVCS construction cost, which is related to its size. The first item in Eq. (21) describes the annual investment cost of BESSs and the second item is the BESS's degradation cost. Eq. (22) indicates the annual investment cost of SDG and Eq. (23) is the calculation of energy losses cost. Note that all the capital recovery factors in Eqs. (18)-(22) can be obtained by Eq. (25), where n^{life} denotes the lifespan of all the infrastructures we mentioned above.

Eq. (24) represents the SEV annual waiting cost. In order to provide satisfactory SEV service, the user's waiting time should be as short as possible. The SEV system's operation model is as follows: a user could look for a SEV with minimum battery level (denoted as c_{rs}) that can support him/her to travel from the original location r to the destination s . If there are multiple SEVs having battery levels greater than c_{rs} , it is assumed that this customer will be assigned to the SEV with lowest battery capacity among all of them in the same SEVCS. If such an SEV is not available at the customer's current location, then the customer is required to wait in a queue until an SEV with c_{rs} is available at his/her location. Once the customer has finished his/her trip in s , the SEV will be assigned to the nearest charging location to park. v in Eq. (24) denotes the unit waiting cost and q_{rs} represents the queuing length of those who want to travel from r to s , which can be obtained by Eq. (26).

The customer queue's length can be determined by: (a) the queue's length in the last time interval, i.e., $q_{rs}(t-1)$; (b) the current user demand from r to s , i.e., $\sum_{q \in Q_{rs}} \beta q_q^{rs}(t)$; and (c) the number of SEVs departing from location r to s . $d_{rs}(c, t)$ in Eq. (26) is the departure rate of SEVs that travel from r to s and are with the battery capacity greater than c .

As shown in Eq. (27), $d_{rs}(c, t)$ is calculated based on: (a) the number of available SEVs with battery level larger or equal to c ; (b) the number of arriving users; (c) the roadway's capacity; and (d) the queue's length. C represents the maximum travel distance when the EV is fully charged. ψ_{rs} is the maximum rate of SEV's traveling on path rs . $d_{rk}(c, t)$ is the departure rate of SEV's traveling from r to k ($k \neq r \neq s$); the distance of the path rk is smaller than that of the path rs . The first item in Eq. (27) means no SEVs with battery capacity less than c_{rs} are able to complete a trip from r to s ; the second item means the number of departing SEVs with battery capacity greater than c_{rs} destined to location $s \neq r$. This number can be determined by: (a) the number of available SEVs at location r with battery capacity greater than c in time t , i.e., $I_r(c, t)$; (b) the total SEVs arriving at location r with the battery capacity larger than c at time t , i.e., $A_r(c, t)$; (c) the SEV demand generated from location r to location s in time interval t , i.e., $\beta q_q^{rs}(t)$; (d) the average user queue's length from location r to location s in interval $t-1$, i.e., $q_{rs}(t-1)$; and (e) the sum of the departing SEVs in interval t that travel to the locations $k \neq s \neq r$, i.e., $\sum_{k \in \Omega^T} d_{rk}(c, t)$. The value of $A_r(c, t)$ and $I_r(c, t)$ in Eq. (27) can be calculated based on Eqs. (28) and (29), respectively. In Eq. (29), $I_r(c, t)$ is determined by available and arrived SEVs at location r based on the previous time interval ($I(\cdot, t-1)$ and $A(\cdot, t-1)$) and the SEVs departure rate in the previous time interval.

The constraints of the proposed model are as below:

(1) Constraints of power flow equality:

$$P_{i,t}^S = \sum_{k \in \Omega^C} P_{k,t}^{CH} + P_{i,t}^{SL} + P_{i,t}^D - \sum_{w \in \Omega^{SDG}} P_{w,t}^{SDG} + \sum_{m \in \Omega^{BE}} P_{ch,m,t} \eta_{ch} - \sum_{m \in \Omega^{BE}} P_{dis,m,t} \eta_{dis} + U_{i,t} \sum_{j \in N^D} U_{j,t} (G_{ij}(x_{ij}) \cos \theta_{ij,t} - B_{ij}(x_{ij}) \sin \theta_{ij,t}) \quad (30)$$

$$Q_{i,t}^S = Q_{i,t}^D + U_{i,t} \sum_{j \in N^D} U_{j,t} (G_{ij}(x_{ij}) \sin \theta_{ij,t} - B_{ij}(x_{ij}) \cos \theta_{ij,t}) \quad (31)$$

(2) Power output and energy capacity constraints of SDGs. Solar power generation of the w th SDG and its corresponding energy capacity should satisfy following constraints:

$$0 \leq P_{w,t}^{SDG} \leq P_{rated}^{SDG} \quad (32)$$

$$0 \leq E_w^{SDG} \leq E_{max}^{SDG} \quad (33)$$

(3) Constraint of SDG's capacity limitation:

$$\sum_{w \in \Omega^{WDG}} E_w^{SDG} \leq \psi S_{load} \quad (34)$$

where S_{load} represents the overall load demand and ψ denotes the ratio of total SDG capacity and load demand.

(4) Constraint of the reversed power flow (f_i). f_i should be limited since it may increase the system loss and lead to overheating of feeders:

$$0 \leq f_i \leq f_i^{max} \quad (35)$$

(5) Constraints of bus voltage and line current:

$$I_{min} \leq I_{ij} \leq I_{max} \quad \forall ij \in \Omega^L \quad (36)$$

$$U_i^{min} \leq U_{i,t} \leq U_i^{max} \quad \forall i \in \Omega^D \quad (37)$$

(6) SEVCS size constraint:

$$z^{min} \leq z_k \leq z^{max} \quad \forall k \in \Omega^C \quad (38)$$

(7) Radiation constraint of SEVCSs:

$$\sum x_{ij} = m^D - m^S \quad \forall ij \in \Omega^L \quad (39)$$

(8) Capacity constraint of the substation:

$$(P_{i,t}^S)^2 + (Q_{i,t}^S)^2 \leq (S_{i,old}^S + y_i S_{i,new}^S)^2 \quad \forall i \in \Omega_{old}^S, \Omega_{new}^S \quad (40)$$

(9) Constraint of feeder's apparent power capacity:

$$(P_{ij,t})^2 + (Q_{ij,t})^2 \leq 0.8 (x_{ij} S_{ij}^L)^2 \quad \forall ij \in \Omega_{old}^L, \Omega_{new}^L \quad (41)$$

$$P_{ij,t} = x_{ij} [-U_{i,t} U_{j,t} (g_{ij} \cos \theta_{ij,t} + b_{ij} \sin \theta_{ij,t}) + U_{i,t}^2 g_{ij}] \quad \forall ij \in \Omega^L \quad (42)$$

$$Q_{ij,t} = x_{ij} [-U_{i,t} U_{j,t} (g_{ij} \sin \theta_{ij,t} - b_{ij} \cos \theta_{ij,t}) - U_{i,t}^2 b_{ij}] \quad \forall ij \in \Omega^L \quad (43)$$

(10) Constraint of BESS's charging/discharging power:

$$0 \leq P_{ch,m}, |P_{dis,m}| \leq P_{bat,max} \quad (44)$$

(11) Constraints of BESS's energy transition and capacity:

$$E_{m,t+1}^{BE} = E_{m,t}^{BE} + (P_{ch,m,t} \eta_{ch} - P_{dis,m,t} \eta_{dis}) \Delta t \quad (45)$$

$$E_{min}^{BE} \leq E_m^{BE} \leq E_{max}^{BE} \quad (46)$$

IV. SIMULATION STUDY

A. Problem Solving Algorithm

The proposed joint ADN planning model is a mixed-integer, non-convex optimization problem over a finite horizon. In this study, Natural Aggregation Algorithm (NAA) [22] is used to solve the model. NAA mimics the biological self-aggregation intelligence and use heuristic rules to do random searching in a high dimensional problem space.

All decision variables listed in Table I are sequentially encoded as individuals in the NAA, indicating a solution of the ADN planning model. The SEV charging profiles calculated in Section II are inputted into the proposed planning together with the solar power data. NAA is then used to solve the proposed model. In this simulation, the maximum iteration number and population size of NAA are set to 800 and 500, respectively.

B. Simulation Description

The proposed ADN model is validated on a coupled system consisting of a 54-node distribution system [23] and a 25-node transportation network [24]. The voltage limit of each node is set as $\pm 5\%$. The SEVCS size upper and lower limits are set to 6 and 12, respectively. The BESS charging and discharging efficiency are set to be 0.9. The power of slow charging for each charging post is 3kW. The charging power of each SEVCS charging facility is set as 25 kW. The SEV model we consider in this study is Nissan Leaf, which is with battery capacity of 30 kWh, power consumption of 0.28 kWh/km, and maximum mileage of 105km [25]. Values of the SEV's unit-time waiting cost v is set to be 20US\$. 8,000 households are assumed to live in this area, and each has 1.86 vehicles [26]. Based on this, SEV penetration rate is set to 9%. μ_s and σ_s for start charging time are set to 18 and 4, respectively. Based on [12], for the daily mileage, μ_d and σ_d are set to 5.0 and 0.75 respectively.

The planning horizon is set to be five years. Four substations are considered in this study with two substations that can be reinforced and two new candidate substations. The construction and maintenance unit costs for all the infrastructures (SEVCSs, feeders, and substations) are set following [10]. The penetration level of both SDGs and BESSs are set to 20%. The investment and maintenance cost of SDG are set to 3,500\$/kW and 0.01\$(/kWh), respectively. The capacity upper limit of SDG is set to 1,000kW; The solar radiation data [27] is converted to the solar power with the conversion efficiency of 0.9. The BESS we considered in this study is based on the Lead-Acid batteries with parameters given in [28].

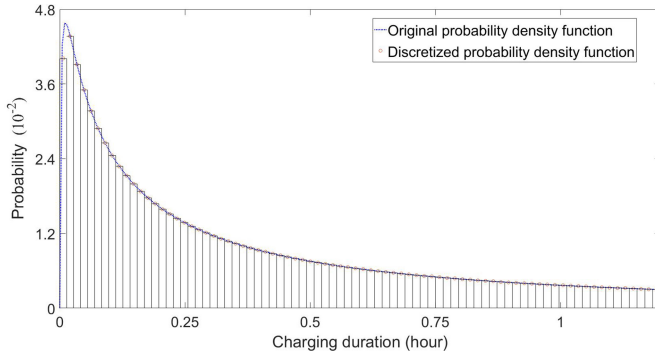


Fig. 3. Discretized probabilistic modeling of charging duration.

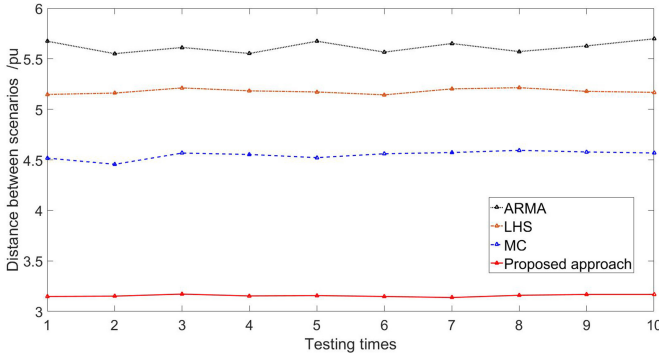


Fig. 4. Comparison results of scenario generation under four methods.

C. Validation of the Stochastic Modelling for SEV Charging Demand

Probabilistic modeling of the uncertain SEV charging load is performed based on the procedures presented in Section II. The result of discretized probability function and optimal discrete quantiles (with $Q = 80$ and $r = 15$) for SEV started charging time and charging duration are shown in Figs. 2 and 3, respectively. In order to investigate the scenario approximation accuracy of the Wasserstein based scenario generation method, the proposed approach is compared with ARMA (Auto Regressive and Moving Average), MC (Monte Carlo) and LHS (Latin Hypercube Sampling) algorithms. Each algorithm forms a scenario set W of t_{ss} , containing 1000 scenarios. Then the distance $D = \sum_{u^i \in Y} p_i \min_{u^j \in W} d(u^i, u^j)$ is used to evaluate each algorithm's performance. Y is the base scenario set which contains 10^6 scenarios sampled by MC from the probability distribution of t_{ss} . $d(u^i, u^j)$ represents the distance between the scenario u^i and u^j ; p_i is the probability of scenario u^i . The value of D is proportional to the approximation error of the original uncertainty problem. The result of ten times tests between all these scenario generation methods are shown in Fig. 4. It can be seen that the proposed Wasserstein based method shows a significant improvement in approximation accuracy over other algorithms.

Based on the generated scenarios from proposed Wasserstein method, the K -medoids ($K = 5$) clustering and Cartesian product fusion method are used to reduce the generated scenarios of joint uncertainty states into five

TABLE II
SOLUTION COMPARISON RESULTS

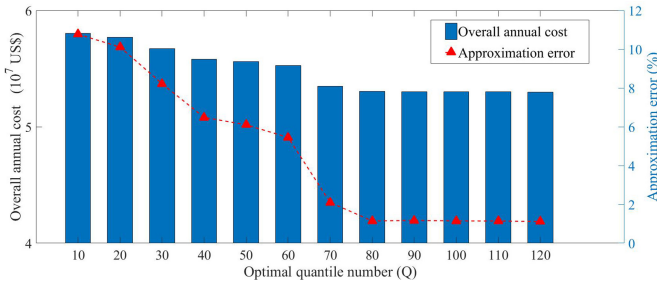
Reduction method	FFS		BR		Proposed method
Case No	Case 1	Case 2	Case 3	Case 4	Case 5
MCS Sampling number	2000	4000	2000	4000	N/A
Scenario Construction Time (s)	643	2954	793	3313	67
Overall annual cost (10^7 US\$)	5.724	5.478	5.692	5.455	5.293
Approximation Error	9.27%	4.52%	8.59%	4.12%	1.07%
Total computation time (s)	2107	4453	2257	5781	1541

representative scenarios. The performance of the developed method is compared with two commonly used methods: FFS (Fast Forward Selection) and BR (Backward Reduction). The results are shown in Table II. Cases 1-4 are based on BR and FFS algorithms with different MCS sampling numbers. Case 5 is the proposed approach described in Section II. The third row in Table II reports the total time of generated scenarios of the SEV daily charging profile. It is clear that when MCS sampling time increases, the computation time for both BR and FFS is increased dramatically, while that of the developed approach is only 67 seconds. This is mainly because the proposed method discretizes the continuous functions of start charging time and charging duration, which can avoid large-scale MCS sampling.

Also, for MCS based FFS and BR algorithms, each scenario is required to compare to all the potential scenarios in original scenario set, resulting in high computational cost. In contrast, the developed scenario reduction approach is based on parallel architecture which the scenarios from each sub-interval are not relevant; this makes the computational performance improved significantly.

For each case, the scenarios of EV charging profile are inputted into the planning model to obtain the approximated total cost. The "approximation error" is the error between: a) the "actual overall annual cost" that is calculated from the EV charging load scenarios construct based on 10^6 MCS and BR, and b) overall annual cost calculated from different cases. From Table II, the approximation error for Cases 1-4 is decreased if we increase the size of the initial scenario set. However, it can be seen that the error of the proposed approach is only 1.07%, which is still smaller than the FFS and BR method even if the initial scenario set contains 4,000 scenarios. In addition, the proposed approach only costs 1,541 seconds to perform computation, indicating that it cannot only achieves high prediction accuracy, but also greatly improve the computational efficiency.

We test the effect of different numbers of optimal discrete quantiles on the approximation error. The results shown in Fig. 5 demonstrate that the increase of Q would decrease the approximation error, and the error tends to be unchanged when the value of Q is 80. The error decreases when the value of Q is larger, but this would also increase the computational cost since the joint state ρ_r would increase exponentially. Therefore, $Q = 80$ is used in this study. In addition, different values of the " r " order are tested, ranging from 1 to 1000. Our experiment results show that the selection of optimal discrete quantiles and their corresponding probabilities are almost

Fig. 5. Sensitivity study of optimal discrete quantiles Q .TABLE III
RESULT OF THE ADN PLANNING FOR DIFFERENT CASES(10⁷ U.S.\$)

Objectives	Case A	Case B	Case C
Overall cost	5.803	5.383	5.293
SEV waiting cost	1.244	0.972	0.921
Total cost of the investment and energy losses	4.559	4.416	4.372
Investment	0.372	0.775	1.173
Maintenance	0.057	0.113	0.164
Production	4.021	3.445	2.968
Energy losses	0.105	0.083	0.067

unchanged when r is larger than 15 (i.e., the variation is less than 0.0001%). Based on this, $r = 15$ is chosen in this study.

D. Simulation Results of Joint ADN Planning

To evaluate the impact of different DERs in the ADN expansion planning, the proposed model is designed with three different cases as follows:

Case A: Consider ADN expansion planning and the construction of SEVCSs;

Case B: Consider ADN expansion planning and construction of both SEVCSs and SDGs;

Case C (the proposed method): Consider ADN expansion planning and construction of all the DERs (SEVCSs, SDGs, and BESSs).

The graphical topology obtained for cases A-C are shown in Fig. 6. The number attached with each SEVCS represents its size. Table III shows the cost items of the obtained solution for each case. Fig. 6(a) and Fig. 6(b) show that the topology of Cases A and B differ in feeder connecting nodes 10, 23, 26, 32, 40, and 48. More SEVCSs are installed in Case B, indicating that the integration of SDGs allows more SEV charging demands to be served. Table III shows that the SEV queuing waiting cost of Case B is decreased by 21.81% compared to Case A due to the improved charging capacity of SEVCSs.

In Case A, due to the SEV charging, the additional transformers are required in nodes 51 (16.7 MVA) and 52 (16.7 MVA), and two new substations are built in nodes 53 (32.2 MVA) and 54 (32.2 MVA). In contrast, for Case B, no substation expansion is needed in node 51, and less transformer capacity in node 52 is needed (13.2 MVA). By installing SDGs and additional SEVCSs in Case B, we can

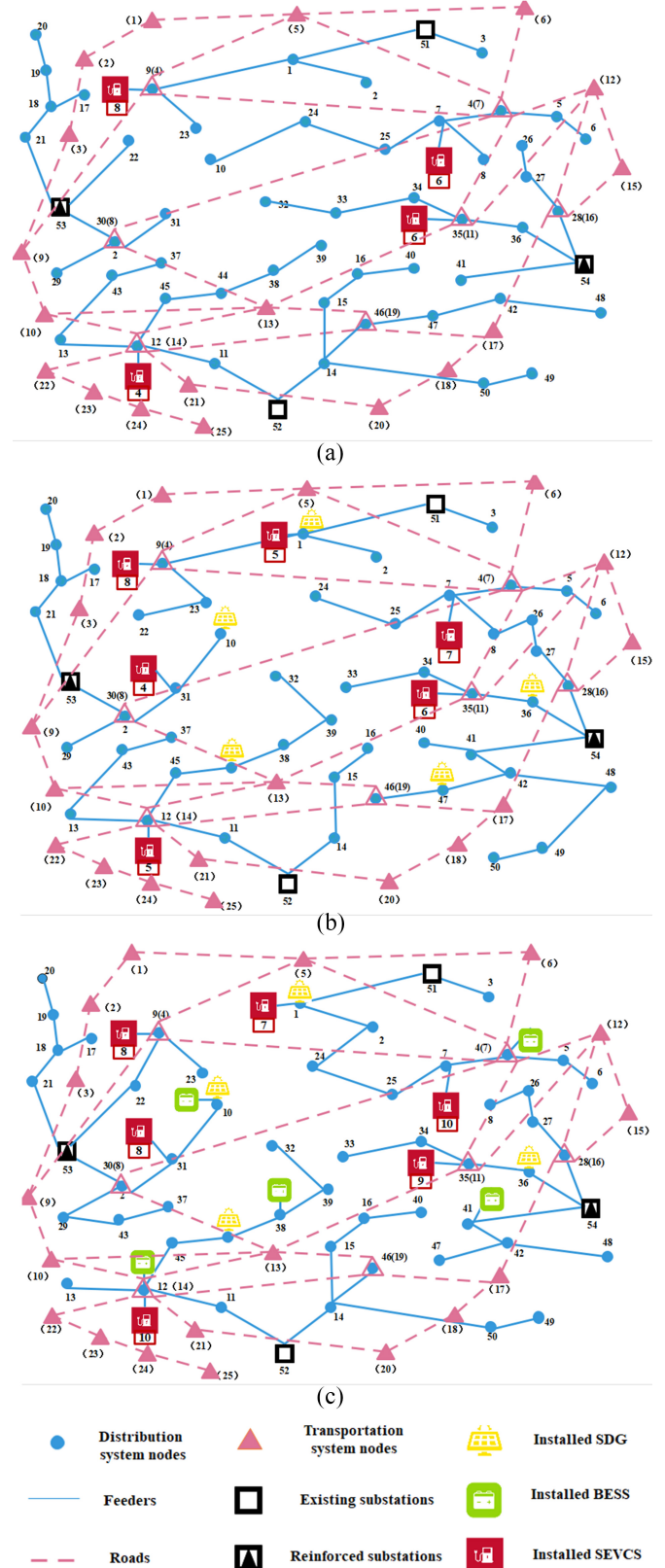


Fig. 6. Graphical topology of the final planning scheme of the coupled distribution and transportation systems (a) Case A; (b) Case B; (c) Case C.

see in Table III that the cost of total investment and maintenance is much higher than those in Case A. However, such cost increase can be mainly offset by decreasing the cost of energy production and loss. It should be pointed out that the

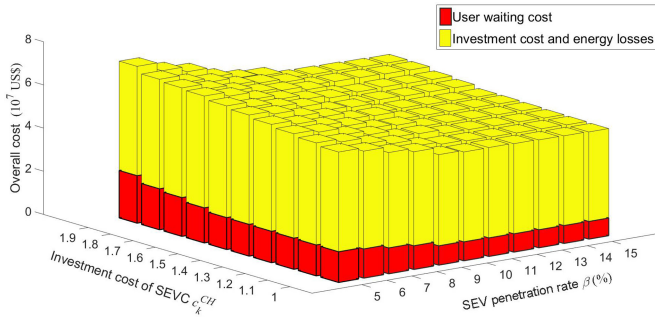


Fig. 7. Sensitivity study of parameters c_k^{CH} and β .

total cost of Case B is reduced by 7.23% compared to that of Case A.

For the proposed model (Case C), when considering BESSs, a notable difference on the planning result is observed. For Case C, SEVCSs' sizes are increased, and the charging capacity is thus improved. As shown in Table III, the integration of BESSs reduces the waiting cost of SEV by 25.96% and 5.24%, compared with Cases A and B, respectively. Presence of BESS makes the investment and maintenance cost increase by 51.32% and 45.13% respectively, compared with Case B; however, the energy loss cost is reduced by 19.27%. This is mainly because the BESS charging and discharging process can vary the distribution of the network power flow and substations' power output. Moreover, the cost of the energy production is decreased by 13.84% since the power can be delivered from the BESSs to grid. Fig. 6(b) and (c) demonstrate the selected locations of SDG and SEVCS in Case C are similar with those in Case B and there is no extra transformer are needed. The overall cost in Case C is reduced by 8.78% and 1.67% compared with Cases A and B, respectively. Therefore, we can conclude that the coordinated integration of BESSs and SDGs in the proposed expansion planning model can not only save the overall cost of investment and energy loss, but also improves the efficiency of SEV utilization.

E. Sensitivity Study

Sensitivity analysis is conducted for the SEV penetration rate (β) and SEVC unit fixed cost (c_k^{CH}). The results are reported in Fig. 7. Fig. 7 shows that when the waiting cost drops from 1.42×10^7 US\$ to 0.894×10^7 US\$ with the increase of 5% SEV penetration rate to 9% penetration rate; then, the waiting cost does not vary too much when β is larger than 9%. This indicates that the waiting cost is not significantly affected by the number of SEV when β is larger than 9%. With the proportional increase of the SEVC's fixed unit cost, Fig. 7 demonstrate that with increase of c_k^{CH} , the customer's waiting cost increases, but the system investment cost and energy loss does not increase too much. This could be explained as that when c_k^{CH} increases, the system tends to sacrifice the user's benefit by increasing the user's waiting time to decrease the number of built stations; as a result, the adverse effect due to the increase of c_k^{CH} can be reduced.

F. Algorithm Validation

To validate the efficiency of NAA on solving the proposed model, we compare NAA with two commonly used heuristic

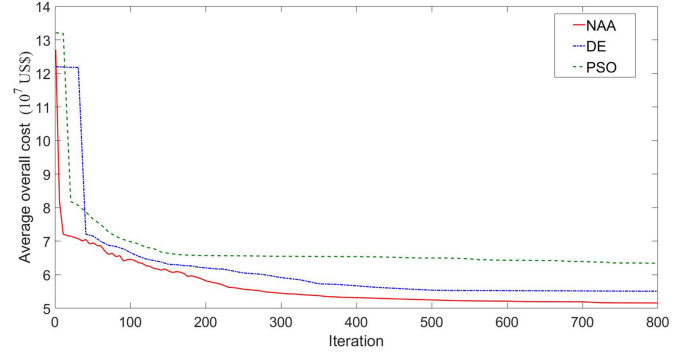


Fig. 8. Algorithm efficiency comparison.

optimization algorithms: Particle Swarm Optimization (PSO) and Differential Evolution (DE). All three algorithms are set with the same population size and maximum iteration time. For fair comparison purpose, five optimization tests were carried out for each algorithm, and the average results are compared. The convergence of the three algorithms on solving the proposed model are shown in Fig. 8. Fig. 8 shows that NAA and DE are significantly better than PSO since the PSO is trapped in the local optima with the increase of iteration. Both DE and NAA continue to find better solutions in the iteration process, but NAA slightly outperforms DE.

V. CONCLUSION

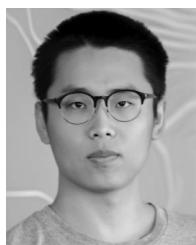
A stochastic ADN planning model is proposed in this study, aiming to optimally determine the location and capacity of SEVCSs, SDGs, BESSs, and the network expansion scheme in a distribution network. It minimizes the sum of total investment cost, energy loss, and SEV waiting cost. The queuing model of shared EV system is introduced to measure the quality of the SEV service. The simulation result validates the effectiveness of the proposed model in terms of improving the SEV's charging service quality and reducing the energy cost. In addition, the simulation also demonstrates the reasonability and necessity of coordinately considering multiple types of energy resources in distribution network planning.

In future, this study can be extended by considering the reliability indices of the coupled ADN system and SEV system. Another possible direction is to implement a dynamic pricing scheme for a large-scale EV-sharing network by considering station-level demand forecasting and vehicle-to-grid integration.

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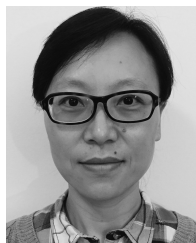


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